

Table 1. Dataset-specific base configurations and boundary statistics.

Dataset	Boundary configuration	Avg. segments	Train max	Val max	Base SumR
ActivityNet	cosine; prom. -1.0 ; dist. 1; smooth 1	36	212	224	183.9
TVR	cosine; prom. 0.2; dist. 2; smooth 1	21	75	52	232.6
Qvhighlights	cosine; prom. 1.3; dist. 2; smooth 3; length regularization	72	143	139	243.5

Table 2. Shared boundary hyperparameters.

Hyperparameter	U1	U2
Multi-scale windows	{2, 4, 8, 16}	
Scale weights	{0.4, 0.3, 0.2, 0.1}	
Peak selection / normalization	prominence / MAD	
MAD prominence	0.4	0.2
Minimum distance / smoothing	1 / 1	

Table 3. Retrieval performance for base, U1, and U2 boundaries.

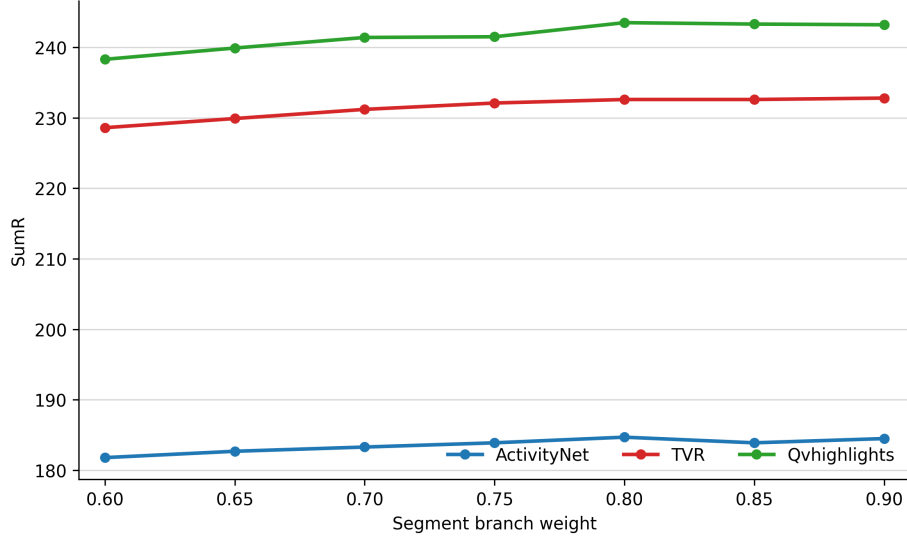
Dataset	Setting	Avg. segments	R@1	R@5	R@10	R@100	SumR
ActivityNet	Base	36	15.0	36.4	49.8	82.8	183.9
ActivityNet	U1	18	14.5	36.2	49.3	83.0	183.1
ActivityNet	U2	19	14.5	36.6	49.7	82.9	183.7
TVR	Base	21	26.0	51.8	62.5	92.3	232.6
TVR	U1	19	26.4	51.7	62.7	92.0	232.8
TVR	U2	21	26.6	51.4	62.8	92.3	233.1
Qvhighlights	Base	72	26.5	54.9	66.8	95.3	243.5
Qvhighlights	U1	139	24.5	54.3	67.0	96.7	242.4
Qvhighlights	U2	149	26.8	54.5	67.7	96.5	245.5

Table 4. Effects of IQS aggregation and Qvhighlights length regularization.

Dataset	Setting	R@1	R@5	R@10	R@100	SumR
ActivityNet	Hard	15.0	36.4	49.8	82.8	183.9
ActivityNet	Soft	14.5	36.3	49.6	83.2	183.6
TVR	Soft	26.0	51.8	62.5	92.3	232.6
TVR	Hard	26.9	51.7	62.3	92.0	233.0
Qvhighlights	Soft	26.5	54.9	66.8	95.3	243.5
Qvhighlights	Hard	24.2	53.4	65.5	96.5	239.6
Qvhighlights	no length regularization	25.8	55.2	66.0	96.1	243.0

Table 5. SumR sensitivity to segment/frame fusion weights.

Segment:Frame	ActivityNet SumR	TVR SumR	Qvhighlights SumR
0.60:0.40	181.8	228.6	238.3
0.65:0.35	182.7	229.9	239.9
0.70:0.30	183.3	231.2	241.4
0.75:0.25	183.9	232.1	241.5
0.80:0.20	184.7	232.6	243.5
0.85:0.15	183.9	232.6	243.3
0.90:0.10	184.5	232.8	243.2

**Figure 1.** SumR across segment/frame fusion weights.**Table 6.** Boundary construction time (s).

Dataset	Wall-clock time (s)
ActivityNet	29.981
TVR	53.653
Qvhighlights	47.766